

Brandon H. Schlomann

My research program works to identify cellular and molecular processes that drive variability in immune responses to infection using live imaging, genetics, and mathematical modeling.

Postdoctoral Fellow, Garcia Lab
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Positions

2020 - present Postdoc University of California, Berkeley
Advisor: Hernan Garcia

Education

2020 Ph.D. Physics, University of Oregon
Advisor: Raghuveer Parthasarathy
Dissertation: Learning biophysical rules of gut bacterial communities
through live imaging of zebrafish
2013 B.A. Physics, with Honors, University of Montana
Advisor: Andrew Ware

Awards and Fellowships

2020 – 2023 James S. McDonnell Foundation Postdoctoral Fellowship
2020 Weiser Doctoral Thesis Award
2016 – 2019 NIH Molecular Biology and Biophysics Training Grant, U. Oregon
2016 NSF GRFP Honorable Mention
2014 First Year Academic Award, Institute of Theoretical Science, U. Oregon
2009 Cal-Murphy Scholarship, U. Montana

First-Author Research Papers (* = equal contributors)

6. Elizabeth Eck*, Bruno Moretti*, **Brandon H. Schlomann***, Jordão Bragantini, Merlin Lange, Xiang Zhao, Shruthi VijayKumar, Guillaume Valentin, Cristina Loureiro, Daniele Soroldoni, Loïc A. Royer, Andrew C. Oates, Hernan G. Garcia. *Single-cell transcriptional dynamics in a living vertebrate*. *bioRxiv* (2024) doi: 10.1101/2024.01.03.574108

5. **Brandon H. Schlomann**, Raghuveer Parthasarathy. *Gut bacterial aggregates as living gels*. *eLife* (2021) 10:e71105

Highlighted in:

Journal Club for Condensed Matter Physics: “Some bacteria more than sum of parts”

4. Travis J. Wiles*, **Brandon H. Schlomann***, Elena S. Wall, Raghuveer Parthasarathy, Karen Guillemin. *Swimming motility of a gut bacterial symbiont promotes resistance to intestinal expulsion and enhances inflammation*. *PLoS Biology*, 18(3): e3000661
3. **Brandon H. Schlomann***, Travis J. Wiles*, Elena S. Wall, Karen Guillemin, Raghuveer Parthasarathy. *Sublethal antibiotics collapse gut bacterial populations by enhancing aggregation and expulsion*. *Proceedings of the National Academy of Science* (2019), 166, 43, 21392-21400

Highlighted in:

Nature Reviews Microbiology, In Brief: “Sublethal antibiotics and a sticky situation”

Nature Reviews Microbiology, Under the Lens: “Light sheets unveil host–microorganism interactions”

Trends in Microbiology, Spotlight: “Bacterial Aggregation Leads to Collective Elimination”

We discovered a mechanism by which sublethal concentrations of antibiotics can have dramatic impacts on bacteria in the gut—namely, by shifting their behavior from a planktonic, motile state to a 3D aggregate phase, in which they are more susceptible to being cleared from the gut by peristaltic fluid flows. This mechanism may be relevant to exposure of antibiotics through environmental contamination or to the use of low-dose antibiotics in agriculture.

2. **Brandon H. Schlomann***, Travis J. Wiles*, Elena S. Wall, Karen Guillemin, Raghuveer Parthasarathy. *Bacterial cohesion predicts spatial distribution in the zebrafish intestine*. *Biophysical Journal* (2018), 115, 22712277
1. **Brandon H. Schlomann**. *Stationary moments, diffusion limits, and extinction times for logistic growth with random catastrophes*. *Journal of Theoretical Biology* (2018), 454, 154-163

Other Research Papers

3. Jarrett F. Leboy, **Brandon H. Schlomann**, Catherine D. Robinson, Brendan J.M. Bohannan. *Phenotypic parallelism during experimental adaptation of a free-living bacterium to the zebrafish gut*. *mBio* (2020) 11, 4, e01519-20
2. Travis J. Wiles, Elena S. Wall, **Brandon H. Schlomann**, Edouard A. Hay, Raghuveer Parthasarathy, Karen Guillemin. *Modernized tools for streamlined genetic manipulation and comparative study of wild and diverse proteobacterial lineages*. *mBio* (2018), 9, 5, e01877-18
1. Travis J Wiles, Matthew Jemielita, Ryan P Baker, **Brandon H Schlomann**, Savannah L Logan, Julia Ganz, Ellie Melancon, Judith S Eisen, Karen Guillemin, Raghuveer Parthasarathy. *Host gut motility promotes competitive exclusion within a model intestinal microbiota*. *PLoS Biology* (2016), 14(7): e1002517

Review Articles

1. **Brandon H. Schlomann** and Raghuveer Parthasarathy. *Timescales of gut microbiome dynamics*. *Current Opinion in Microbiology* (2019) 50:56–63

Workshops organized

5. 3rd Physics of Life Symposium, Chan-Zuckerberg Biohub, September 25, 2024
4. Modeling the Evolutionary Dynamics of Cheating in Natural Viral Infections, Santa Fe Institute, April 10-15, 2024
3. 2nd Physics of Life Symposium, Chan-Zuckerberg Biohub, October 12, 2023
2. Slow-Fast Dynamics in Biological Systems, Santa Fe Institute October 5-9, 2023
1. Slow-Fast Dynamics in Biological Systems, Santa Fe Institute March 23-28, 2023

Conference Talks and Seminars

11. *Deciphering spatial patterns of immune response in fruit fly larvae*, American Physical Society March Meeting, Minneapolis, MN, March 7, 2024
10. *Coarse-graining immune responses*, CZ Biohub, Physics of Life Symposium, January 23, 2023
9. *Kinetics and statistics of gut bacterial cluster sizes*, University of British Columbia, SBME Seminar, September 24, 2020
8. *Antibiotics and aggregation in the zebrafish gut* (with Raghuvier Parthasarathy), Transport in the digestive tract: experiments, modeling, applications to microbiology, Paris, France, October 22, 2020
7. *Live imaging of gut bacterial communities in zebrafish: spatial structure, fast dynamics, and antibiotic perturbations*, 2nd International Fish-Microbiota Workshop, Eugene, OR, September 4, 2019
6. *Live imaging of gut bacterial communities in zebrafish: spatial structure, fast dynamics, and antibiotic perturbations*, PacNow Quantitative Biology Meeting, Portland OR, August 22, 2019
5. *Bacterial cohesion predicts spatial distribution in the zebrafish intestine*, American Physical Society March Meeting, Boston, MA, March 5, 2019
4. *Swim/stick tradeoffs for gut bacterial symbionts*, American Physical Society March Meeting, Los Angeles, CA, March 7, 2018
3. *Catastrophe and fast dispersal dynamics of the zebrafish gut microbiota following sublethal antibiotics*, International Fish-Microbiota Workshop, Trondheim, Norway, June 20, 2017
2. *Visualizing the response of a gut bacterial population to antibiotic perturbations*, American Physical Society March Meeting, New Orleans, LA, March 13, 2017
1. *Response dynamics of gut bacterial communities*, META Center Symposium, Eugene, OR, June 16, 2016

Students mentored

5. Hana Mir (Summer 2023), UC Berkeley Physics graduate student. *I mentored Hana during her rotation project on theoretical studies of nonlinear wave generation in biochemical models of immune signaling.*
4. Jazmin Sandhu (March 2023 - present), UC Berkeley Physics undergraduate student. *I mentored Jazmin in her experimental studies of fruit fly immune responses, as part of the PI2 program.*

3. Tiffany (Ting-Wei) Pai (September 2022 - present), UC Berkeley Molecular and Cell Biology undergraduate student. *I mentored Tiffany in experimental studies of fruit fly immune responses, as part of the SURF program*
2. Genesis Ferrer Imbert (July 2022 - September 2022), UC Berkeley Biophysics rotation student. *I mentored Genesis during her rotation project in experimental studies of fruit fly immune responses and computational image analysis.*
1. Sathvik Anantakrishnan (June 2021 - February 2022), IIT Madras Biological Sciences Masters student. *I remotely mentored Sathvik in theoretical and computational studies of the zebrafish segmentation clock and coupled oscillators. Sathvik is currently a Biophysics PhD student at UC Berkeley.*

Teaching, Outreach, DEI

8. UC Berkeley Quantitative Biology Bootcamp (2024). *I led experimental rotations for an 8 day summer course introducing incoming biology graduate students to quantitative biology.*
7. UCSF TRAIN UP Inclusive Mentoring Course (2023). *I attended this course on lab management and mentorship.*
6. UC Berkeley POWER (Physical Sciences Opportunities for Womxn in Education and Research) program mentor (2023-2024). *I mentored two community college students and helped them prepare for transferring to a 4-year university.*
5. UC Berkeley Quantitative Biology Bootcamp (2023). *I led experimental rotations for an 8 day summer course introducing incoming biology graduate students to quantitative biology.*
4. UC Berkeley Quantitative Biology Bootcamp (2022). *I was a teaching assistant for an 8 day summer course introducing incoming biology graduate students to quantitative biology.*
3. Summer Academy to Inspire Learning, University of Oregon (2014–2018). *I led small group demonstrations for high school students during an annual summer outreach program.*
2. PHYS 204/205/206, University of Oregon (2014–2015). *I taught algebra-based physics lab.—*
1. Physics Study Jam, University of Montana (2007–2009). *I worked at a drop-in tutoring center, tutoring undergraduate students in various physics courses.*